

Econ 30065: Microeconomics

Sara Negrelli

Monopoly

Topics

- Market power
- Monopoly pricing
- Welfare effects of monopoly pricing

Market Power

- **Market power:** when a firm can profitably charge a price that is above its marginal cost ($P > MC$)
 - **Monopoly market:** a market with a single seller
 - **Oligopoly market:** a market with a few sellers
 - In order for a firm to have market power, competition cannot be intense on that market
- Determining if a market is a monopoly can be trickier than simple definitions might suggest
 - Need to define the relevant market, which includes close substitutes

How do firms become monopolists?

- Government awards and patents (cable TV companies in local communities, drug patents)
- Innovation (iPod)
 - Erode over time unless continuous innovation
- Ownership of all of an essential input (De Beers diamond producer in the 80s)
- Other firms do not find the market profitable (ex: economies of scale)

Monopoly Pricing

- Monopolist will choose the price (and quantity) that maximizes its profit (total Revenue – total Cost), given the demand for its product
 - Optimality condition $MR=MC$
 - If marginal revenue exceeded marginal cost the firm would be better off selling more
 - If marginal revenue were less than marginal cost the firm would be better off selling less
- Marginal cost curve applies as usual
- But now firm is not price taker (as in perfect competition): $MR \neq P$
 - Need to examine the shape of the marginal revenue curve: additional revenue from selling one extra unit of the good

Marginal Revenue

- Increase in sales quantity, from $Q - \Delta Q$ to Q changes revenue in two ways
 - **Output expansion effect:** sell ΔQ additional units, each at price of $P(Q)$
 - **Price reduction effect:** increased sales quantity requires a reduction in price from $P(Q - \Delta Q)$ to $P(Q)$ for all the previous units

Marginal Revenue

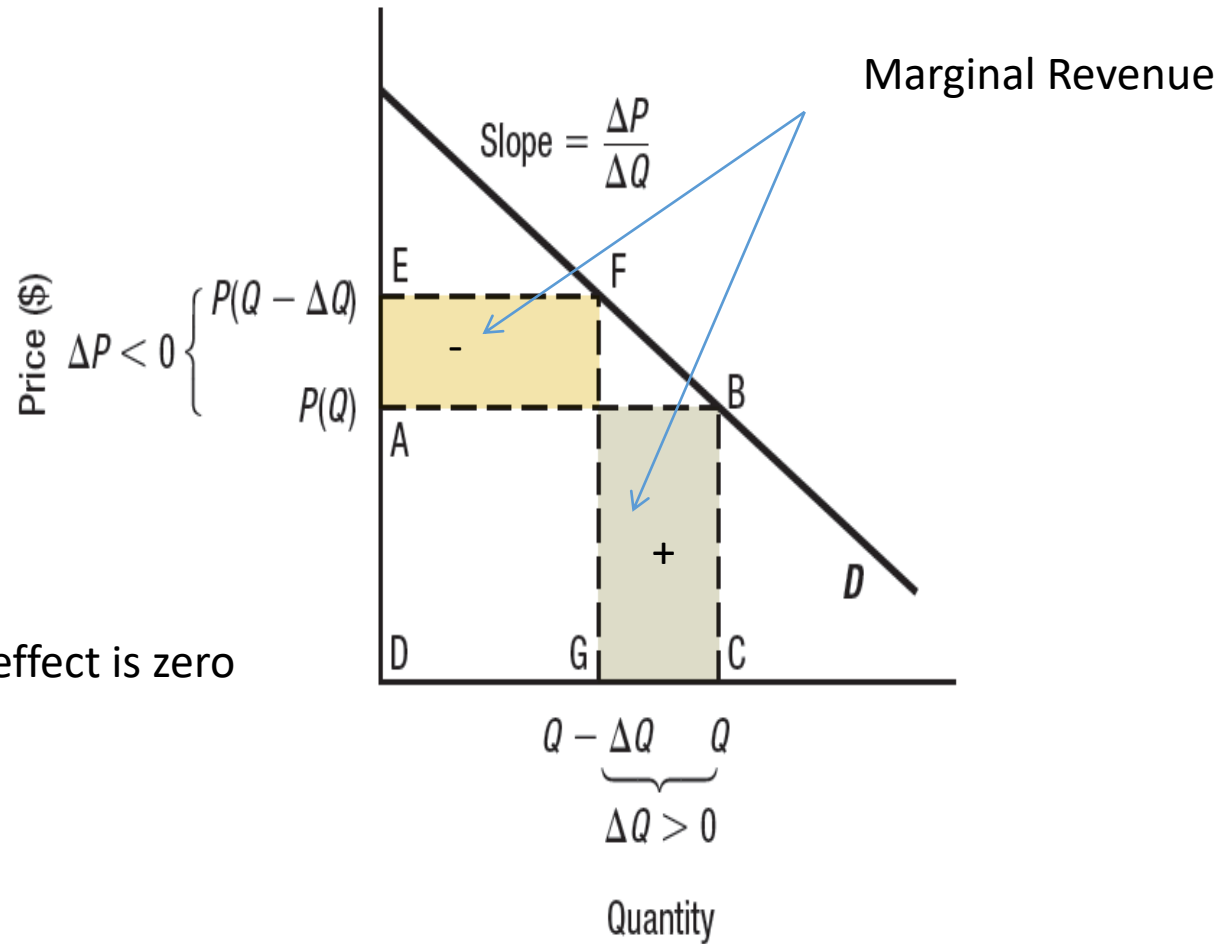
(b) Downward-sloping demand curve

$$MR = P(Q) + \left(\frac{\Delta P}{\Delta Q}\right)Q$$

↑
Output
Expansion
Effect

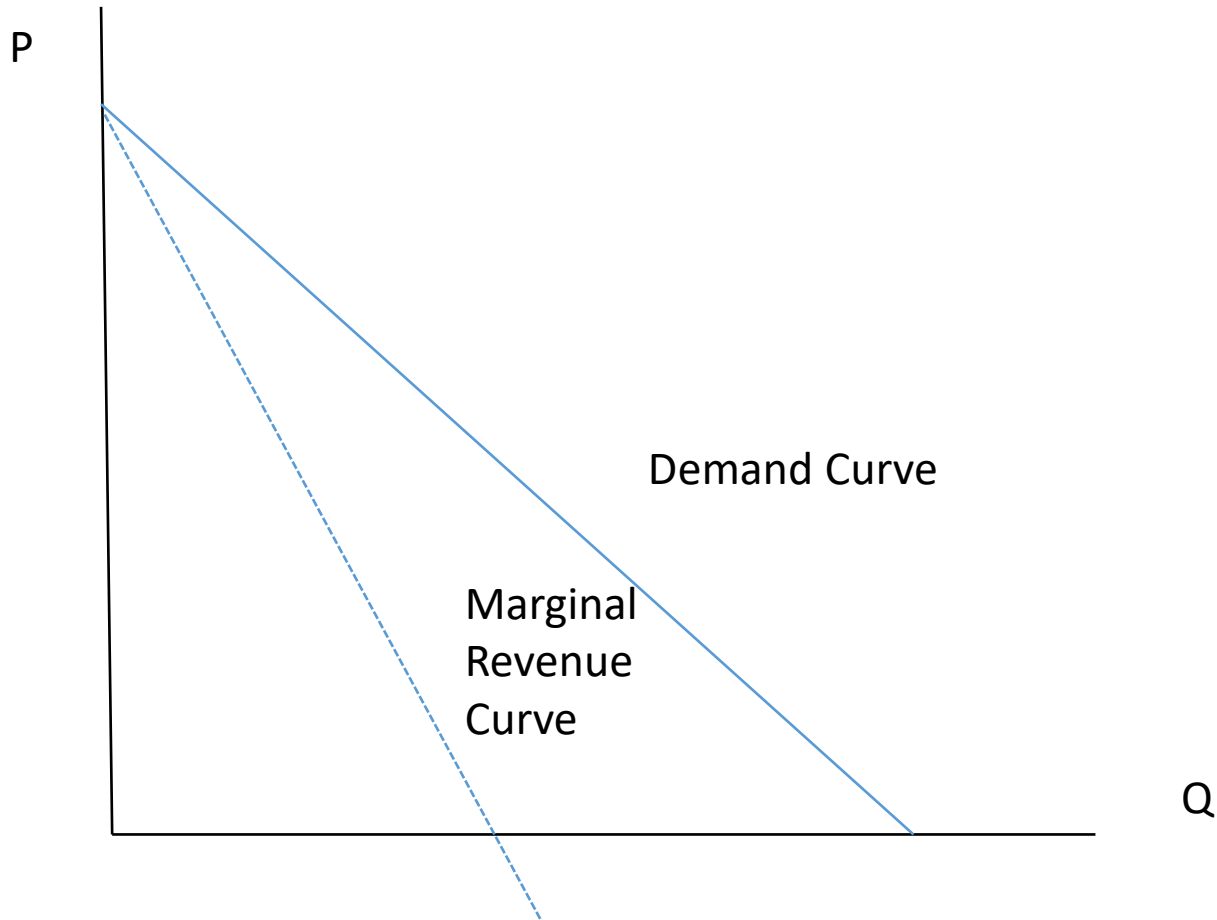
↑
Price
Reduction
Effect

- In Perfect competition price reduction effect is zero (price takers) → $MR=P$
- In Monopoly it is negative → $MR<P$



Marginal Revenue and Demand Curves

- Marginal Revenue Curve crosses the X-axis halfway between the origin and the demand curve
- MR is downward sloping because, as we increase Q , the price reduction effect (negative) becomes greater than the output expansion effect (when do the 2 effects have the same size?)



Monopolist profit maximization

➤ **Step 1: Quantity rule**

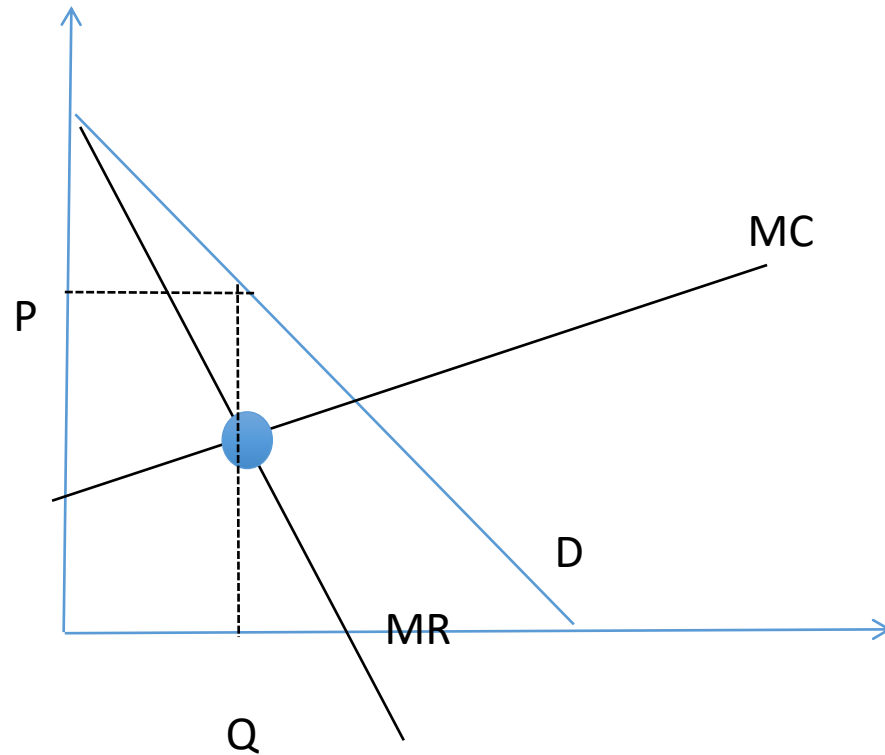
- Identify positive sales quantities at which $MR=MC$. If more than one positive sales satisfies the conditions, identify the one that produces the highest profit.

➤ **Step 2: Shut-down rule**

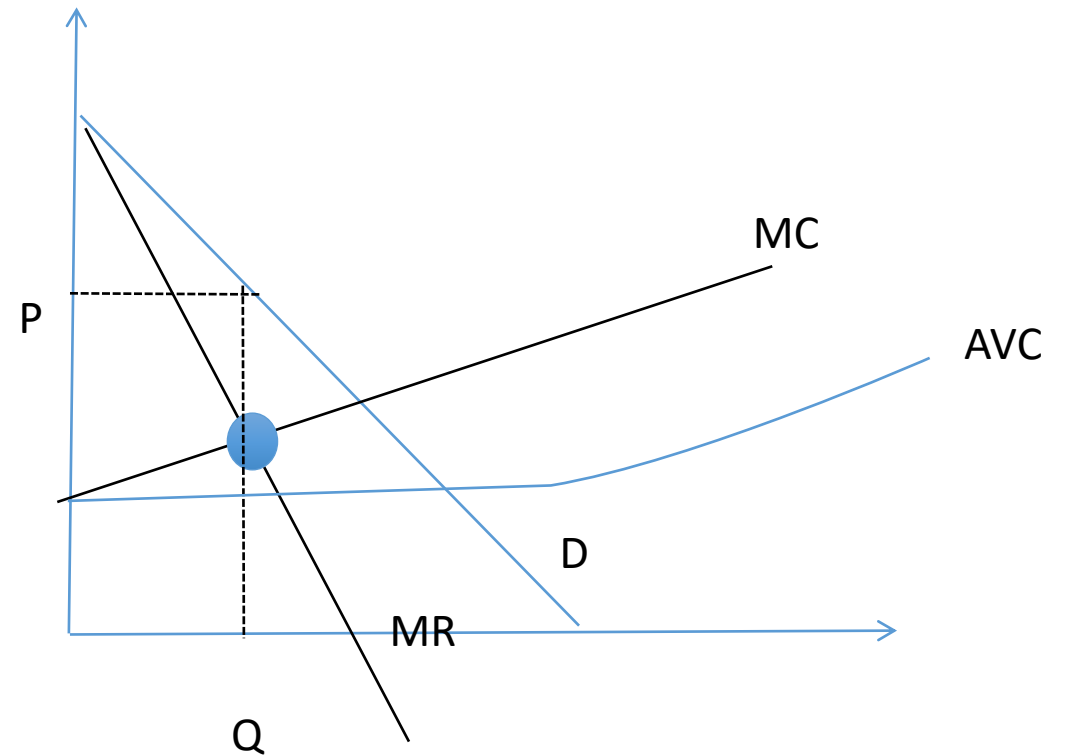
- Check whether the most profitable positive sales quantity from Step 1 is greater profit than shutting down. If it is, that is the profit maximizing choice. If not, then selling nothing is the best option
 - $TR \geq \text{Variable Costs}$
 - $p \geq AVC(q)$
 - $p \geq \min AVC$

Profit Maximizing Price and Sales Quantity

Step 1



Step 2



Example

Consider a monopolist whose annual demand function is $Q^D = 16000 - 200P$. Suppose that it has annual variable costs of $VC = 20Q + 0.00125Q^2$ and $MC = 20 + 0.0025Q$. In addition it has a fixed cost of 48000\$ per year. What is his profit maximizing P and Q ?

Solution

Monopoly $\rightarrow$ $MR=MC$

1) Find $TR=P \times Q$

$$TR=(80-Q/200)Q$$

$$MR=80-Q/100$$

2) Set $MR=MC$:

$$80-Q/100=20+0.0025Q \rightarrow 60=0.0125Q \rightarrow Q=4800$$

$$P=56$$

3) Check shut down rule: $TR=4800 \times 56=268800$

Variable costs=124800 $\rightarrow$ stay open and produce!

Markup: A Measure of Market Power

- In perfect competition $P=MC$
 - firm has no market power
- A monopolist facing a downward sloping demand curve will set its price above marginal cost
 - Extent to which price exceeds marginal cost is a measure of monopolist's market power
 - Usually economists measure this difference as a percentage of a firm price
 - A firm's **markup, price-cost margin**, or **Lerner index** equals the difference between its price and its marginal cost, as a percentage of its price:
$$\frac{P - MC}{P}$$
- We translate the optimality condition that marginal revenue should equal marginal cost into a rule of thumb that shows us market power

Markup: A Measure of Market Power

$$\frac{P - MC}{P} = -\frac{1}{E^d}$$

- Monopolist's markup at its profit-maximizing price always equals the reciprocal of the elasticity of demand, times negative one
- The less elastic the demand curve, the greater the firm's markup over its marginal cost
- When demand is less elastic, raising the price is more attractive because fewer sales are lost

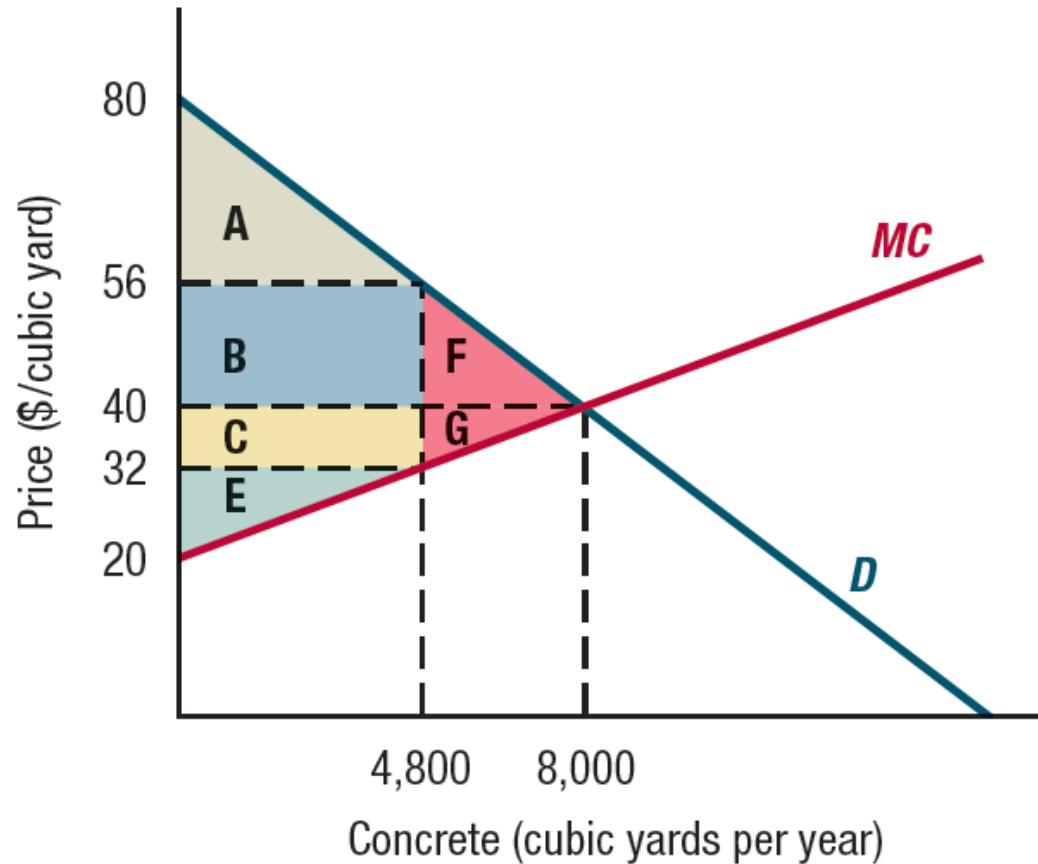
Welfare Effects of Monopoly Pricing

- By charging a price above marginal cost, the monopolist makes consumers worse off than under perfect competition
 - Consumers who buy the product pay more for it
 - Some who would have bought it under perfect competition will not buy it at the higher price
- Welfare effects of monopoly pricing:
 - Firm gains
 - Consumers lose
 - Deadweight loss incurred
- Deadweight loss from monopoly pricing is the amount by which aggregate surplus falls short of its maximum possible level, which is attained in a competitive market

Computing welfare effects

- Compare surplus in Monopoly with surplus in perfect competition
- Surplus in Monopoly:
 - $0.5Q[(D_{\text{intercept}} - S_{\text{intercept}}) + (P - MC(Q))]$ –avoidable fixed costs
- Surplus in perfect competition:
 - Find P and Q in perfect competition: set $P = MC$, where P is the inverse demand function → find Q, and then P
 - Compute surplus: $0.5Q(D_{\text{intercept}} - S_{\text{intercept}})$ –avoidable fixed costs
- Subtract surplus in monopoly from surplus in PC → find DWL!

Welfare Effects of Monopoly Pricing



CS in monopoly: A
PS in monopoly: B+C+E (-
avoidable fixed costs)
DWL in monopoly: F+G

Example continued

Compute the DWL of monopoly:

➤ In monopoly $P=56$, $Q=4800$

➤ In perfect competition instead: $P=MC$:

$$P=20+0.0025Q$$

$$80-0.005Q=20+0.0025Q \rightarrow Q=8000, P=40$$

➤ $DWL=0.5(4800-8000)(56-32)=38400$

Example 2

A monopolist is facing the following inverse market demand:

$P = 50 - Q$, where P is the price and Q is the quantity sold in the market. Monopolist's short run costs are equal to

$$TC = 20Q$$

- Compute marginal revenue and marginal cost.
- Draw them, together with the market demand curve, on a graph.
- Compute the equilibrium quantity and price.

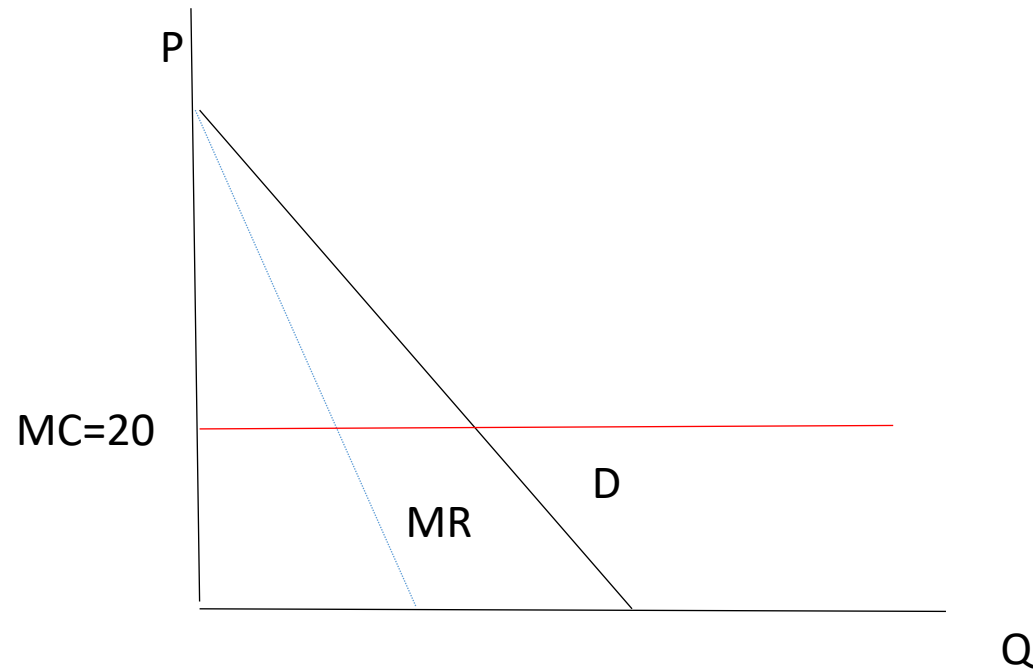
Example 2

$$MC=20$$

$$MR=50-2Q$$

To find optimal Q and P set $MR=MC$:

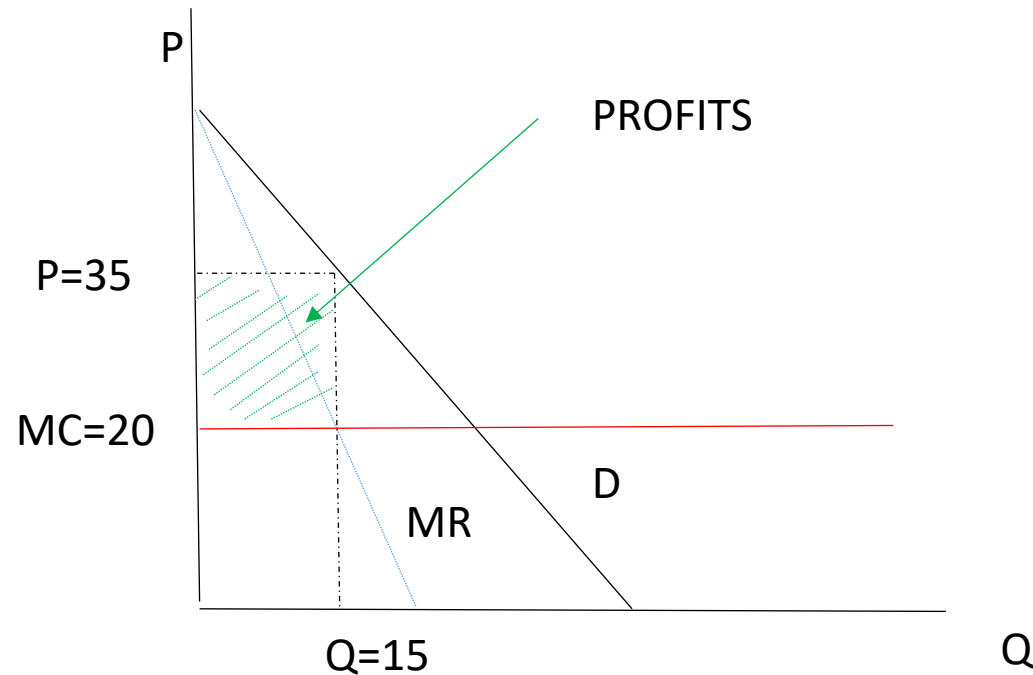
$$50-2Q=20 \rightarrow Q=15, P=35$$



Example 2

➤ Calculate monopolist's profits (Π) and represent them on the previous graph.

➤ $\Pi^* = TR - TC = 35(15) - 20(15) = 225$



Example 2

- Calculate demand price elasticity at the monopoly equilibrium price:
 - $(P-MC)/P = -1/E_d$
 - $E_d = -2.33$
- Assuming that the profits of the monopolist are as calculated above, how will the equilibrium price and quantity change if a fixed amount tax of 100 is introduced?
 - *A lump sum tax (fixed amount tax) does not modify optimality conditions for the monopolist. Therefore, the equilibrium quantity and price would not change; however, monopolist's profits will go down by the amount of tax:*
$$\Pi' = \Pi^* - T = 125$$